

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - VII

Roll No. :S046	Name:Abhishek Yadav
Class :BSC IT	Batch : 01
Date of Assignment :2/9/2026	Date/Time of Submission :2/9/2026

```
import matplotlib.pyplot as plt
```

A. Write a Python program to create a scatter plot of Age and Height of 5 students.

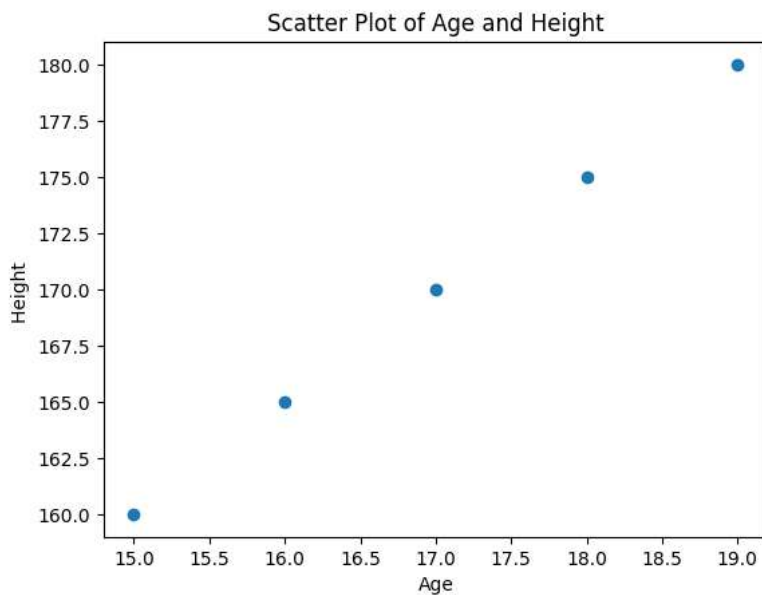
```
import numpy as np

age = [15, 16, 17, 18, 19]
height = [160, 165, 170, 175, 180]

plt.scatter(age, height)

plt.xlabel("Age")
plt.ylabel("Height ")
plt.title("Scatter Plot of Age and Height")

plt.show()
```



B. Write a Python program to create a bar chart showing marks of 5 students.

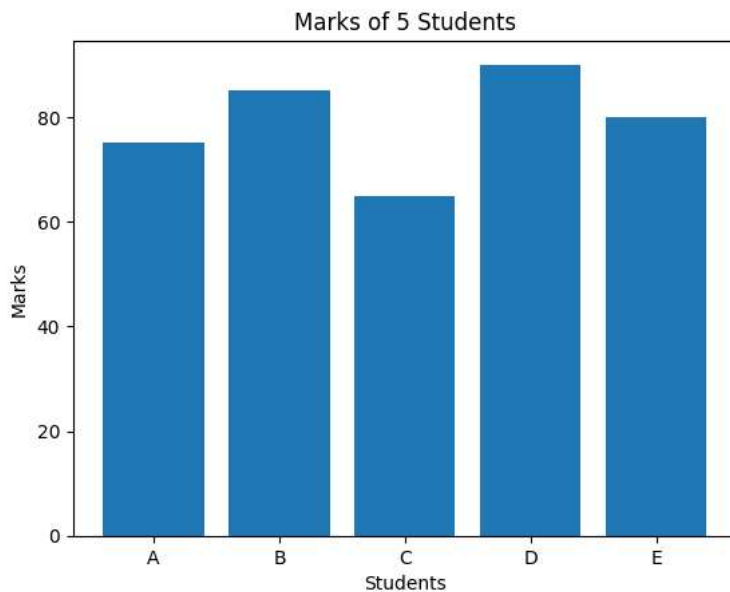
```
import numpy as np

students = ["A", "B", "C", "D", "E"]
marks = [75, 85, 65, 90, 80]

plt.bar(students, marks)

plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Marks of 5 Students")

plt.show()
```



C. Write a Python program to create a histogram showing the distribution of marks of 10 students. **bold text**

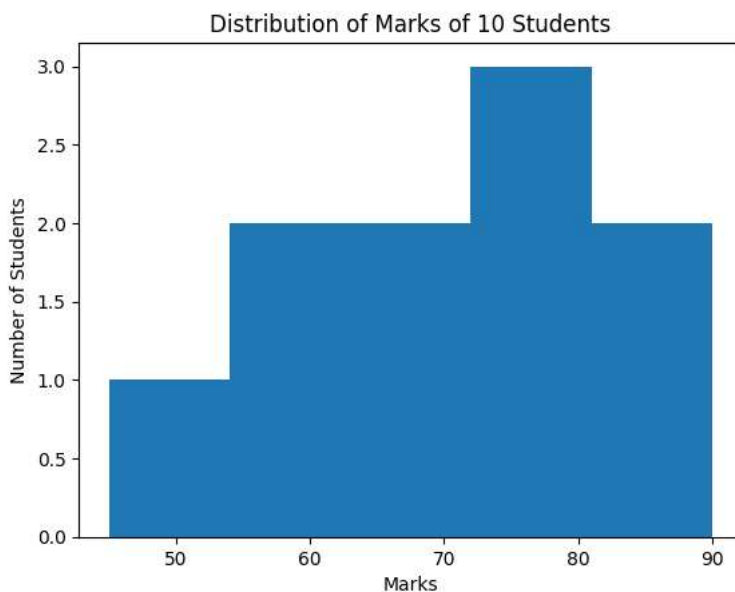
```
import numpy as np

marks = [45, 55, 60, 65, 70, 72, 75, 80, 85, 90]

plt.hist(marks, bins=5)

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Marks of 10 Students")

plt.show()
```



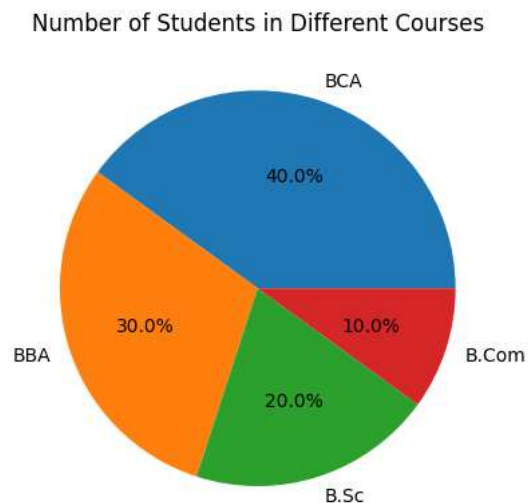
D. Write a Python program to create a pie chart showing the number of students in different courses. **bold text**

```
import numpy as np

courses = ["BCA", "BBA", "B.Sc", "B.Com"]
students = [40, 30, 20, 10]

plt.pie(students, labels=courses, autopct="%1.1f%%")
```

```
plt.title("Number of Students in Different Courses")  
plt.show()
```



E. Write a Python program to create a bar chart showing sales of 5 different products.

```
import numpy as np  
  
products = ["Product A", "Product B", "Product C", "Product D", "Product E"]  
sales = [50, 70, 40, 90, 60]  
  
plt.barh(products, sales)  
  
plt.xlabel("Products")  
plt.ylabel("Sales")  
plt.title("Sales of 5 Different Products")  
  
plt.show()
```

